

BSDS (2025 Semester II): Statistics II

Quiz 1

NAME :

Solution Set

ROLL :

Time: 60 minutes

Total marks: 30

1. The number of COVID infected people in a certain community having N people is K_0 (K_0 is unknown and N is extremely large). Let $\theta = K_0/N$ be the proportion of infected people. To estimate θ , a person takes $n = 50$ simple random samples with replacement from this community.

- (a) Employ an appropriate model to characterize this population. Justify your answer.
(b) Based on your assumed model, find the UMVUE of θ .

(a) Let $X_1, \dots, X_n$ be the SRS of size $n=50$. [5 + 5]

For any $i=1, \dots, n$,

$$X_i = \begin{cases} 1 & \text{if the } i\text{th sample is infected} \\ 0 & \text{otherwise.} \end{cases}$$

$$P(X_i=1) = \frac{\text{\# infected persons}}{\text{population size}} = \frac{K_0}{N} = \theta$$

So, $X_i \sim \text{Bernoulli}(\theta)$ for any $i=1, \dots, n$

Further, for a simple random sample with replacement

$$P(X_i=x, X_j=y) = P(X_i=x) P(X_j=y)$$

$$+ x \in \{0,1\} \quad y \in \{0,1\}$$

So, $X_i \stackrel{iid}{\sim} \text{Bernoulli}(\theta)$; $i=1, \dots, n$; is the model.

(b) The joint pmf of $X_1, \dots, X_n$ given the model is

$$f_{\theta; x_1 \dots x_n}(\mathbf{x}) = \theta^{\sum x_i} (1-\theta)^{n-\sum x_i} \\ = \exp \left\{ \sum_{i=1}^n x_i \log \left(\frac{\theta}{1-\theta} \right) + n \log (1-\theta) \right\}$$

So, the joint distribution belongs to the exponential family, and $\sum_{i=1}^n X_i$ is a complete-sufficient statistic (CSS) of θ .

$$\text{Now, } E\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n E(X_i) = n\theta,$$

as $E(X_i) = 1 \cdot P(X_i=1) = \theta$ and $X_1, \dots, X_n$ are iid.

$$\text{So, } E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \theta.$$

Thus, $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ is a function of a CSS and is unbiased for θ . Hence $\bar{X}_n$ is the UMVUE of θ .

2. Let $X_1, \dots, X_n$ be a random sample from $\text{Exponential}(\theta)$ distribution, with common p.d.f.

$$f_X(x) = \theta \exp\{-\theta x\}, \quad x > 0, \quad \theta > 0.$$

Consider three estimates of $g(\theta) = \sqrt{\theta}$, $T_1(\mathbf{X}) = n^{-1} \sum_{i=1}^n X_i^{-1/2}$, and $T_2(\mathbf{X}) = (\bar{X}_n)^{-1/2}$.

(a) Is $T_{1,n} = T_1(\mathbf{X})$ unbiased for $g(\theta)$? If not, can you find a function h_1 such that $h_1(T_{1,n})$ is unbiased for $g(\theta)$?

$$(a) \quad E(T_1(\mathbf{X})) = \frac{1}{n} \sum_{i=1}^n E(X_i^{-1/2}) = E(\bar{X}_1^{-1/2}).$$

$$\text{Now,} \quad E(\bar{X}_1^{-1/2}) = \theta \int_0^{\infty} x^{-1/2} e^{-\theta x} dx.$$

Let $u = \theta x$ then $du = \theta dx$; $x = \frac{u}{\theta}$ and

x	0	∞
u	0	∞

$$\text{So, } E(\bar{X}_1^{-1/2})$$

$$= \int_0^{\infty} u^{-1/2} \theta^{1/2} e^{-u} du$$

$$= \sqrt{\theta} \Gamma_{1/2} = \sqrt{\pi \theta}$$

So, $\frac{1}{\sqrt{\pi}} T_1(\mathbf{X})$ is unbiased for $\sqrt{\theta}$; and $T_1(\mathbf{X})$ is NOT unbiased for $\sqrt{\theta}$. Here, $h_1(x) = \frac{x}{\sqrt{\pi}}$.

(b) Is $T_{2,n} = T_2(\mathbf{X})$ unbiased for $g(\theta)$? If not, can you find a function h_2 such that $h_2(T_{2,n})$ is unbiased for $g(\theta)$? [Hint: You may use the additive property of Exponential distribution.]

$$T_2(\underline{x}) = T_{2,n} = \frac{1}{\sqrt{\bar{x}_n}} = \frac{\sqrt{n}}{\sqrt{\sum_{i=1}^n x_i}} = \frac{\sqrt{n}}{\sqrt{y}},$$

where $y = \sum_{i=1}^n x_i$.

We know that, if $x_i \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$,

then $y = \sum_{i=1}^n x_i \sim \text{Gamma}(n, \lambda)$.

Then pdf of y is

$$f_{X,Y}(y) = \frac{\theta^n}{\Gamma(n)} y^{n-1} e^{-y\theta}; \quad y > 0, \theta > 0.$$

Now, $E(T_2(\underline{x})) = \sqrt{n} E(\bar{y}^{1/2})$

$$= \sqrt{n} \frac{\theta^n}{\Gamma(n)} \int_0^\infty y^{n-3/2} e^{-\theta y} dy$$

Consider the transformn. $w = \theta y$, then

$dw = \theta dy$, $y = w/\theta$ and

y	0	∞
w	0	∞

$$\begin{aligned} \text{So, } E(T_2(\underline{x})) &= \frac{\sqrt{n} \theta^{n-1}}{\Gamma(n)} \int_0^\infty w^{n-3/2} e^{-w} dw \\ &= \frac{\sqrt{n}}{\Gamma(n)} \sqrt{\theta} \Gamma(n-1/2). \end{aligned}$$

So, $T_2(\underline{x})$ is NOT unbiased for $\sqrt{\theta}$, but $h_2(T_2(\underline{x}))$ is unbiased for $\sqrt{\theta}$, where $h_2(z) = \frac{\Gamma(n)}{\sqrt{n} \Gamma(n-1/2)} z$.

- (c) Is any of the two unbiased estimators $h_1(T_{1,n})$ and $h_2(T_{2,n})$ the UMVUE of $\sqrt{\theta}$? Justify your answer. [4 + 5 + 5]

The joint pdf of $X_1, \dots, X_n$ is

$$\begin{aligned} f_{\theta; \underline{x}}(\underline{x}) &= \theta^n \exp \left\{ -\theta \sum_{i=1}^n x_i \right\} \\ &= \exp \left\{ n \log \theta - \theta \sum_{i=1}^n x_i \right\}. \end{aligned}$$

Thus, the joint distributions belong to the exponential family and $Y = \sum_{i=1}^n X_i$ is a complete and sufficient statistic for θ .

Since $h_2(T_2(\underline{x}))$ is a function of Y and is unbiased for $\sqrt{\theta}$, $h_2(T_2(\underline{x}))$ must be the UMVUE.

Further, as UMVUE, if exists, is unique and

$$h_1(T_1(\underline{x})) \neq h_2(T_2(\underline{x})) \text{ with prob. 1,}$$

$h_1(T_1(\underline{x}))$ is NOT the UMVUE.

3. Let X be a random variable with parameters (θ, γ) p.d.f.

$$f_X(x) = \frac{\theta \gamma^\theta}{x^{\theta+1}}, \quad x \geq \gamma, \quad \theta > 0, \gamma > 0.$$

What is the distribution of $Y = \log X - \log \gamma$? Hence or otherwise find the expectation of $\log X$.
[4 + 2]

$$Y = \log \frac{x}{\gamma} \Rightarrow \gamma \exp\{Y\} = x,$$

Jacobian of transformation:

$$J = \left| \frac{\partial x}{\partial y} \right| = \gamma e^y$$

Thus,

$$\begin{aligned} f_{\theta, \gamma; Y}(y) &= \frac{\theta \gamma^\theta}{(\gamma e^y)^{\theta+1}} \gamma e^y \\ &= \frac{\theta \cancel{\gamma}^\theta}{\cancel{\gamma}^\theta} \frac{e^{-\theta y - y + y}}{e} \\ &= \theta e^{-\theta y} \end{aligned}$$

Also, if $x = \gamma$ then $y = 0$ and if $x = \infty$ then $y = \infty$. So, the range of y is $[0, \infty)$.

$\therefore Y \sim \text{Exponential}(\theta)$.

$$E(Y) = \int_0^\infty \theta y e^{-\theta y} dy = \frac{1}{\theta} \int_0^\infty u e^{-u} du = \frac{1}{\theta} \Gamma_2 = \frac{1}{\theta}$$

$$\therefore E[\log X - \log \gamma] = \frac{1}{\theta} \Rightarrow E(\log X) = \underline{\underline{\frac{1}{\theta} + \log \gamma}} \quad [\text{Ans}]$$